SUPPLEMENTARY APPENDIX

Trastuzumab structure and affinity comparison

Publication note (29 October 2026): This online supplement preserves Appendix A1–A8 from the final manuscript. References to the evidence archive or original companion below describe the separately retained research package. The public GitHub repository contains this appendix, its displayed tables and figures, and eight selected numerical CSV files; it does not contain the full evidence archive.

The appendix documents the five-structure mean analysis of 834-residue Fv–HER2 complexes. Reference numbers follow the main manuscript. WT, V1 and V55 denote wild type, Variant 1 and Variant 55. The primary cohort contains 60 complexes; all 68 scored records are retained for provenance. The machine-readable companion provides full IDs, selection flags and calculations.

Numeric display rule. Floating-point results are shown to two decimal places. Nonzero magnitudes below 0.01 use scientific notation with a two-decimal mantissa. Conditional K_D values also use scientific notation when below 0.10 nM or at least 10,000 nM; other conditional K_D values use fixed two-decimal notation. Primary calculations, including Appendix A8, use the retained source scores without additional rounding. The companion retains source inputs, intermediate values, R and RT constants, and provenance hashes; the report-precision comparison is labeled separately.

A1 Sequences and experimental reference

Table A1. Sequence-defined constructs and BLI reference.

Variant	Printed segment VH 96–110	VH substitutions	K_D (nM)	$\Delta\Delta G_{exp}$ (kcal mol ^{1})
WT	CSRWGGDGFYAMDYW	None	0.40	0.00
V1	CSRWKGDGFYEHDYW	G100K, A106E, M107H	0.33	−0.11
V55	CSRYGQPGLYEHDYW	W99Y, G101Q, D102P, F104L, A106E, M107H	18.34	2.27

K_D values are from Mason et al., Supplementary Table 6 [5]. Energy values are calculated only for discussion using 298.15 K, the appendix's exact arithmetic convention corresponding to the main text's rounded 298 K description. WT is the anchor. Substitution numbers refer to sequential chain H positions (source chain B), not an asserted Kabat/IMGT conversion.

WT VH 120 residues

```
EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRY
ADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWGGDGFYAMDYWGQGTLVTVSS
```

V1 VH 120 residues

```
EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRY
ADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRWKGDGFYEHDYWGQGTLVTVSS
```

V55 VH 120 residues

```
EVQLVESGGGLVQPGGSLRLSCAASGFNIKDTYIHWVRQAPGKGLEWVARIYPTNGYTRY
ADSVKGRFTISADTSKNTAYLQMNSLRAEDTAVYYCSRYGQPGLYEHDYWGQGTLVTVSS
```

Unchanged VL 107 residues

DIQMTQSPSSLSASVGDRVTITCRASQDVNTAVAWYQQKPGKAPKLLIYSASFLYSGVPS

RFSGSRSGTDFTLTISSLQPEDFATYYCQQHYTTPPTFGQGTKVEIK

HER2 construct 607 residues

TQVCTGTDMKLRLPASPETHLDMLRHLYQGCQVVQGNLELTYLPTNASLSFLQDIQEVQG

YVLIAHNQVRQVPLQRLRIVRGTQLFEDNYALAVLDNGDPLNNTTPVTGASPGGLRELQL

RSLTEILKGGVLIQRNPQLCYQDTILWKDIFHKNNQLALTLIDTNRSRACHPCSPMCKGS

RCWGESSEDCQSLTRTVCAGGCARCKGPLPTDCCHEQCAAGCTGPKHSDCLACLHFNHSG

ICELHCPALVTYNTDTFESMPNPEGRYTFGASCVTACPYNYLSTDVGSCTLVCPLHNQEV

TAEDGTQRCEKCSKPCARVCYGLGMEHLREVRAVTSANIQEFAGCKKIFGSLAFLPESFD

GDPASNTAPLQPEQLQVFETLEEITGYLYISAWPDSLPDLSVFQNLQVIRGRILHNGAYS

LTLQGLGISWLGLRSLRELGSGLALIHHNTHLCFVHTVPWDQLFRNPHQALLHTANRPED

ECVGEGLACHQLCARGHCWGPGPTQCVNCSQFLRGQECVEECRVLQGLPREYVNARHCLP

CHPECQPQNGSVTCFGPEADQCVACAHYKDPPFCVARCPSGVKPDLSYMPIWKFPDEEGA

CQPCPIN

All chains are numbered sequentially from 1. Source 1N8Z chains B/A/C map to output H/L/A. The template manifest identifies modeled HER2 positions 102–110, 303–305, 361–364 and 581–590 (26 residues). These are study-chain positions; a canonical HER2 numbering map was not supplied. The common HER2 sequence is not the full-length receptor.

Specified disulfide pairs

L23–L88; H22–H96; A4–A31; A140–A170; A173–A182; A177–A190; A198–A205; A202–A213; A214–A222; A218–A230; A233–A242; A246–A273; A277–A289; A293–A309; A312–A316; A320–A345; A453–A482; A489–A498; A493–A506; A509–A518; A522–A538; A541–A554; A545–A562; A565–A574; A578–A601.

A2 Protocol and reproducibility records

ESMFold2 received separate H, L and HER2 entities and retained its predicted three-chain coordinates without a framework assembly transform. Other routes inherited the template orientation and underwent local restrained repair. The archive contains the selected coordinates and scoring execution records, embedded refinement audits for all 53 repaired outputs, and QC/confidence and inference metadata for all 52 ESMFold2 candidates. It does not contain a complete upstream rerun record: referenced template, IgFold and ABodyBuilder3 parent coordinates and original generation invocations remain unrecovered, and 37 unselected ESMFold2 candidates lack archived coordinates. Table A2 and the inventory below distinguish retained records from these limits; absence from the archive does not establish that a record was never generated.

Table A2. Recorded software and model identity.

Stage	Recorded identity and settings
ESMFold2	Python 3.12.3; esm 3.4.1; torch 2.11.0; transformers 4.57.6; biopython 1.88. Loops=20; sampling steps=100; diffusion samples=1; esmc_offload=true. Per-candidate seeds and checkpoint hashes retained; literal inference command not archived.
IgFold parents	Checkpoint families 1, 2, 3, 5, 1 in repair-group order. Original package version, command and weight hashes unavailable.
ABodyBuilder3 parents	pLDDT-loss/base-loss alternate over six repair groups. Base-loss has no pLDDT head. Original package version, command and weight hashes unavailable.
Template parents	FoldX BuildModel runs 1–5, each with variant-specific WT controls. Parent-generation executable identity and original RepairPDB/BuildModel arguments unavailable.
Repair	OpenMM 8.6; CUDA mixed precision; 8 threads. amber14/protein.ff14SB.xml plus implicit/obc2.xml; pH 7; 25 specified disulfides.
FoldX scoring	Version 5.1 verified in scoring logs. AnalyseComplex; analyseComplexChains=HL,A; temperature=298.15; pH=7; water=IGNORE. repair_before_scoring=false.
EvoEF2 scoring	ComputeBinding; split=HL,A; Total binding-interaction energy, kcal mol^{-1} [8]. Executable hash recorded; version/commit unavailable. repair_before_scoring=false.

Repair and stopping procedure

Apply seeded Gaussian coordinate perturbations with SD 0.05 Å to backbone and SD 0.20 Å to side chains, as specified in the embedded refinement settings. Restrain the parent geometry with a harmonic potential using 0.5 k d^2, k = 1000 and 100 kJ mol^{-1} nm^{-2} for backbone and side chains. Minimize with at most 1000 iterations at force tolerance 10 kJ mol^{-1} nm^{-1}, followed by at most 5000 at tolerance 1. Recorded final RMS forces were below 1 kJ mol^{-1} nm^{-1}. Admit only structures meeting QC and per-chain movement limits of 1 Å Cα RMSD and 2 Å heavy-atom RMSD. Absolute movement metrics are recorded relative to the parent; the unarchived parent files are required to recalculate them.

Use repair seeds 9 103 000–9 103 004 for IgFold and template FoldX, and 9 103 000–9 103 005 for ABodyBuilder3. Template groups contain a separate WT–V1 pair and WT–V55 pair. IgFold groups cycle checkpoint families 1, 2, 3, 5, 1; ABodyBuilder3 alternates pLDDT-loss and base-loss. Score QC-accepted structures immediately and stop after the current group once each construct has at least five distinct, doubly scored structures. CPU retries with the same seed are not new observations.

Table A3. Structural admission thresholds.

Check	Strict bound	Exploratory bound
Intraresidue C–C (Å)	1.15–1.80	1.05–1.90
Intraresidue C–N (Å)	1.10–1.75	1.00–1.85
Intraresidue C–O (Å)	1.05–1.65	0.95–1.75
Intraresidue C–S (Å)	1.55–2.05	1.45–2.15
Peptide C–N (Å)	1.10–1.80	1.00–2.00
Specified disulfide SG–SG (Å)	1.70–3.10	Unchanged
Severe nonbonded heavy-atom distance	No pair <1.2 Å	Unchanged
Antibody–HER2 contacts	≥3 residue pairs within 5 Å	Unchanged
Combined framework Cα RMSD	≤4 Å	Unchanged
Sequence and coordinates	Exact coverage; required heavy atoms; finite coordinates	Unchanged

Bonded exclusions apply to clash screening. Short nonbonded contacts below 1.8 Å and disulfide distances above 2.3 Å generate warnings. Strict-tier means compliance with the listed admission checks, not verified biological validity. No native-epitope test,

minimum ipTM, or comprehensive Ramachandran, chirality and peptide-planarity gate was applied. The full framework mask and exclusion implementation were not included in the export.

ESMFold2 selection

The 52-candidate inventory contains WT/V1/V55 counts of 18/17/17; 12/9/10 were QC-eligible. The retained selection script filters on QC pass, orders strict tier first and then descending ipTM, pTM and mean pLDDT with identifier tie-breaking, and greedily keeps models distinct from previously retained models of the same construct. The recorded threshold is at least 0.10 Å in internal Fv RMSD or antigen-aligned Fv-pose RMSD. Selection stops at five per construct; scoring follows selection, and neither energy nor experimental K_D enters the ranking. Among 17 V1 candidates, four passed strict QC (seeds 3, 14, 16 and 15), five passed exploratory QC only and eight failed. Seed9 had the highest ipTM among the additional exploratory-only passes and was retained fifth after the four strict passes. Higher-ipTM seeds 10 and 4 failed the disulfide hard gate. The record-level ranking therefore explains seed9 inclusion. All-candidate geometry and distinctness cannot be independently recomputed without the 37 unarchived candidate coordinate sets and the complete QC implementation. The initial candidate-sampling stop rule remains unrecovered.

Fingerprints and source files

Source 1N8Z SHA-256: ed934d36 503 335c6764b08 880f9e5093 3d99eb13b9bf145 339cfcaee150eb21e

ESMFold2 model.safetensors SHA-256: 138fd4350d6892b81ce6be7ff9bf5a93 ae9d4d3751f46a27 438a3f9f0dcefa0e

FoldX scoring executable SHA-256: 3f892 047bc23 997ae21d71a34c1c5e62 01e394a997 694 587cbe5e66c11 557aa7

EvoEF2 executable SHA-256: 140cf4a0699a4ddda36e50cb03e12c34 e8e033c357d9 099 606e2a7 918 721fade

Repair code SHA-256: 107c6a3575c98e461 897bc52ebb2fbe3 405b690e8328f04 874e7ac6b9aba1bde

1N8Z + FoldX evidence.json SHA-256: ed10beb4ddad2ef323 809df1e4 565 865 f0bf3b40f1e2599a78 553ba0dda026cb

ESMFold2 evidence.json SHA-256: fa39abf525a97 213b87db527 147f3a83 616 051d69db2cf366cc7f7a4a08cd1d3

IgFold evidence.json SHA-256: 8d32d97f9d7094e63f5ba25 905d042d6 f65e57a85da9ddabf79e48a224e64 943

ABodyBuilder3 evidence.json SHA-256: 6b9ea1d6f66 812ff0d739 859 566d4d2c 20d8f02 800 321d5d6f9314c68aaf7fd8

Spaces in the displayed hashes separate two 32-character blocks; concatenate them for verification. Within mason_four_routes_two_scores_20 260 911, each route retains evidence/evidence.json, a selected_scores.csv (ESMFold2) or raw_scores.csv, selected PDBs and evidence/scores/{foldx,evoef2}/RECORD_ID/attempt-001 command/result/program files with scoring inputs. All 60 selected coordinates and 120 selected scoring bundles were located and checked. ESMFold2 metadata retain per-candidate seeds, software versions, inference parameters and checkpoint hashes, including the ESMFold2 model and six ESMC weight shards; weight binaries are not distributed. All 52 candidate QC/confidence summaries are retained, but only 15 selected candidate coordinates are archived. Raw prediction/confidence output files referenced by their hash maps are not archived.

For the three repaired routes, the embedded audits retain Gaussian perturbation settings, force-field names, seeds, software/platform identity and code/input/output hashes. The 12 IgFold, six ABodyBuilder3 and 20 template parent paths referenced across all records are not archived. Original upstream prediction/model-building invocations and relevant checkpoint/version provenance, the matching repair source, and force-field XML hashes remain unrecovered. A separately supplied 1N8Z file matches the recorded source hash and supports Appendix A7; it does not replace those route parents.

A historical export_esm_top5.py script was recovered from the original workspace during this revision and is identified separately in the supplement (SHA-256 96cd0003d5e73 934a69c1525ff086a4c2cff031da608e0b78a4e5c7f4d514bf6). It confirms tier-first ranking, distinctness

filtering, scoring after selection and export of selected coordinate sets. It was not part of the original data package. Its complete imported QC/geometry dependency chain and inference execution history were not recovered; no end-to-end rerun is claimed.

A3 Structural quality and confidence

Table A4. Selected ESMFold2 model quality and confidence.

Model	Rank	Tier	ipTM	pTM	Mean pLDDT	Fv fit (Å)	Contacts	Short
WT s7	1	strict	0.28	0.68	0.86	0.49	67	1
WT s15	2	strict	0.28	0.68	0.86	0.45	57	3
WT s1	3	strict	0.28	0.67	0.86	0.47	49	1
WT s14	4	strict	0.28	0.68	0.86	0.46	89	0
WT s2	5	strict	0.26	0.67	0.85	0.48	84	1
V1 s3	1	strict	0.33	0.73	0.86	0.49	103	2
V1 s14	2	strict	0.30	0.70	0.85	0.50	90	1
V1 s16	3	strict	0.29	0.69	0.86	0.46	103	0
V1 s15	4	strict	0.26	0.68	0.85	0.45	35	1
V1 s9	5	exploratory	0.83	0.88	0.88	0.48	121	0
V55 s0	1	strict	0.32	0.70	0.86	0.56	61	1
V55 s11	2	strict	0.28	0.68	0.85	0.45	85	1
V55 s1	3	strict	0.27	0.68	0.85	0.49	48	1
V55 s12	4	strict	0.27	0.68	0.85	0.53	88	1
V55 s16	5	strict	0.27	0.68	0.85	0.42	56	0

Confidence retains its recorded 0–1 scale. Rank is selection order within construct, not energy rank. Contacts are antibody–HER2 residue pairs within 5 Å; Short counts nonbonded contacts below 1.80 Å. All models have zero severe clashes and complete required heavy atoms. V1 s9 has terminal H120/A607 C–O lengths of 1.0192/1.0095 Å: these fail the strict 1.05–1.65 Å interval but satisfy the exploratory 0.95–1.75 Å interval. Its A202–A213 disulfide distance of 2.3035 Å triggers the 2.30 Å warning, not the 1.70–3.10 Å rejection gate. All other selected models are strict. Framework fit does not establish native epitope recovery.

Table A5. QC ranges across all repaired outputs.

Route	Variant	n	Framework RMSD (Å)	Contacts	Maximum-chain Cα move (Å)
1N8Z + FoldX	WT	10	0.19–0.19	47–49	0.23–0.23
1N8Z + FoldX	V1	5	0.19–0.20	50–52	0.22–0.24
1N8Z + FoldX	V55	5	0.19–0.21	50–52	0.23–0.23
IgFold	WT	5	0.58–0.59	44–56	0.24–0.27
IgFold	V1	5	0.57–0.63	53–61	0.24–0.28
IgFold	V55	5	0.55–0.64	42–52	0.24–0.30
ABodyBuilder3	WT	6	0.70–0.74	47–52	0.23–0.24
ABodyBuilder3	V1	6	0.70–0.77	48–51	0.23–0.24
ABodyBuilder3	V55	6	0.69–0.77	51–52	0.23–0.24

All 53 repaired records are strict-tier, including additional controls/non-distinct outputs. Ranges describe the recorded QC and parent-displacement metrics; parent files are required for independent recomputation. Complete per-record QC, warnings and hashes are in evidence.json. These are restrained conformers, not independent structural determinations.

A4 Raw binding scores and analysis selection

Tables A6–A9 list all 68 scored records. Included denotes membership in the current five-structure mean, not merely QC eligibility or an available WT match. Exactly five WT, five V1 and five V55 records enter each route. For repaired

routes these are the first five entries per construct in the saved ordered eligible list; the ESMFold2 top-five selection is unchanged. No energy or experimental affinity determines inclusion. The selected template WT baseline contains WT–V1 controls from runs 1–3 and WT–V55 controls from runs 1–2; it is shared across both unpaired variant contrasts.

Both scoring functions use kcal mol^{-1}. The revised scoring reports explicitly supply these units and cite the EvoEF2 paper's Figure 5 [8]. The unchanged evidence.json still uses the generic label "EvoEF2 score units"; the companion records that original label for provenance. Reported units do not constitute thermodynamic calibration.

Table A6. 1N8Z + FoldX raw binding energies and inclusion.

Record ID	Construct	Included	FoldX (kcal mol ^{1})	EvoEF2 (kcal mol ^{1})
R01	WT	Yes	−17.01	−58.38
R02	V1	Yes	−17.91	−59.50
R03	WT	Yes	−21.92	−59.89
R04	V55	Yes	−20.78	−59.59
R05	WT	Yes	−16.66	−59.07
R06	V1	Yes	−17.57	−58.92
R07	WT	Yes	−23.35	−60.37
R08	V55	Yes	−23.09	−62.95
R09	WT	Yes	−14.72	−55.43
R10	V1	Yes	−18.34	−56.87
R11	WT	No	−17.97	−54.10
R12	V55	Yes	−15.61	−53.86
R13	WT	No	−16.48	−58.43
R14	V1	Yes	−17.90	−59.35
R15	WT	No	−16.82	−58.49
R16	V55	Yes	−15.86	−59.54
R17	WT	No	−14.15	−56.33
R18	V1	Yes	−14.53	−57.48
R19	WT	No	−14.15	−56.50
R20	V55	Yes	−19.85	−55.84

R identifiers follow this route's scoring report; ESMFold2 labels identify construct and seed. Full IDs, source geometry hashes and current inclusion flags are in raw_scores_all.csv. Excluded records remain supporting evidence and are not zero-valued observations.

Table A7. ESMFold2 raw binding energies and inclusion.

Record ID	Construct	Included	FoldX (kcal mol ^{1})	EvoEF2 (kcal mol ^{1})
WT s7	WT	Yes	39.62	51.31
WT s15	WT	Yes	44.28	33.76
WT s1	WT	Yes	17.49	10.92
WT s14	WT	Yes	33.22	14.24
WT s2	WT	Yes	42.11	59.46
V1 s3	V1	Yes	33.58	46.51
V1 s14	V1	Yes	19.53	17.78
V1 s16	V1	Yes	25.97	38.56
V1 s15	V1	Yes	10.46	74.76
V1 s9	V1	Yes	−1.38	−53.24
V55 s0	V55	Yes	33.28	33.50
V55 s11	V55	Yes	32.44	48.49

Record ID	Construct	Included	FoldX (kcal mol ^{1})	EvoEF2 (kcal mol ^{1})
V55 s1	V55	Yes	10.25	13.17
V55 s12	V55	Yes	62.55	83.61
V55 s16	V55	Yes	9.46	5.51

R identifiers follow this route's scoring report; ESMFold2 labels identify construct and seed. Full IDs, source geometry hashes and current inclusion flags are in raw_scores_all.csv. Excluded records remain supporting evidence and are not zero-valued observations.

Table A8. IgFold raw binding energies and inclusion.

Record ID	Construct	Included	FoldX (kcal mol ^{1})	EvoEF2 (kcal mol ^{1})
R01	WT	Yes	−14.68	−46.34
R02	V1	Yes	−10.09	−50.97
R03	V55	Yes	−7.67	−36.46
R04	WT	Yes	−1.25	−38.97
R05	V1	Yes	−9.30	−48.58
R06	V55	Yes	−9.69	−56.25
R07	WT	Yes	−12.49	−39.65
R08	V1	Yes	−17.40	−42.31
R09	V55	Yes	−14.42	−39.66
R10	WT	Yes	−13.00	−49.34
R11	V1	Yes	−15.23	−55.09
R12	V55	Yes	−6.60	−45.92
R13	WT	Yes	−15.39	−47.54
R14	V1	Yes	−11.58	−54.61
R15	V55	Yes	−10.75	−35.63

R identifiers follow this route's scoring report; ESMFold2 labels identify construct and seed. Full IDs, source geometry hashes and current inclusion flags are in raw_scores_all.csv. Excluded records remain supporting evidence and are not zero-valued observations.

Table A9. ABodyBuilder3 raw binding energies and inclusion.

Record ID	Construct	Included	FoldX (kcal mol ^{1})	EvoEF2 (kcal mol ^{1})
R01	WT	Yes	−16.07	−63.46
R02	V1	Yes	−17.62	−62.08
R03	V55	Yes	−19.58	−67.83
R04	WT	Yes	−14.95	−62.50
R05	V1	Yes	−18.60	−63.55
R06	V55	Yes	−17.86	−66.86
R07	WT	Yes	−15.59	−63.31
R08	V1	Yes	−17.08	−64.86
R09	V55	Yes	−17.34	−68.25
R10	WT	No	−15.06	−62.68
R11	V1	Yes	−18.43	−64.89
R12	V55	Yes	−18.50	−66.45
R13	WT	Yes	−16.68	−64.15
R14	V1	Yes	−16.70	−62.76
R15	V55	No	−17.52	−68.20
R16	WT	Yes	−16.91	−62.54
R17	V1	No	−17.50	−65.81
R18	V55	Yes	−17.57	−66.24

R identifiers follow this route's scoring report; ESMFold2 labels identify construct and seed. Full IDs, source geometry hashes and current inclusion flags are in raw_scores_all.csv. Excluded records remain supporting evidence and are not zero-valued observations.

A5 Mean energies and conditional affinity

The central analysis subtracts the same five-structure WT arithmetic mean from each five-structure variant arithmetic mean, separately within route and scoring method. Parent/seed identifiers remain provenance, not a pairing rule. SDs and observed ranges in Tables A10–A11 describe individual construct energies only. No ΔΔG SD, standard error or confidence interval is inferred from these correlated conformers.

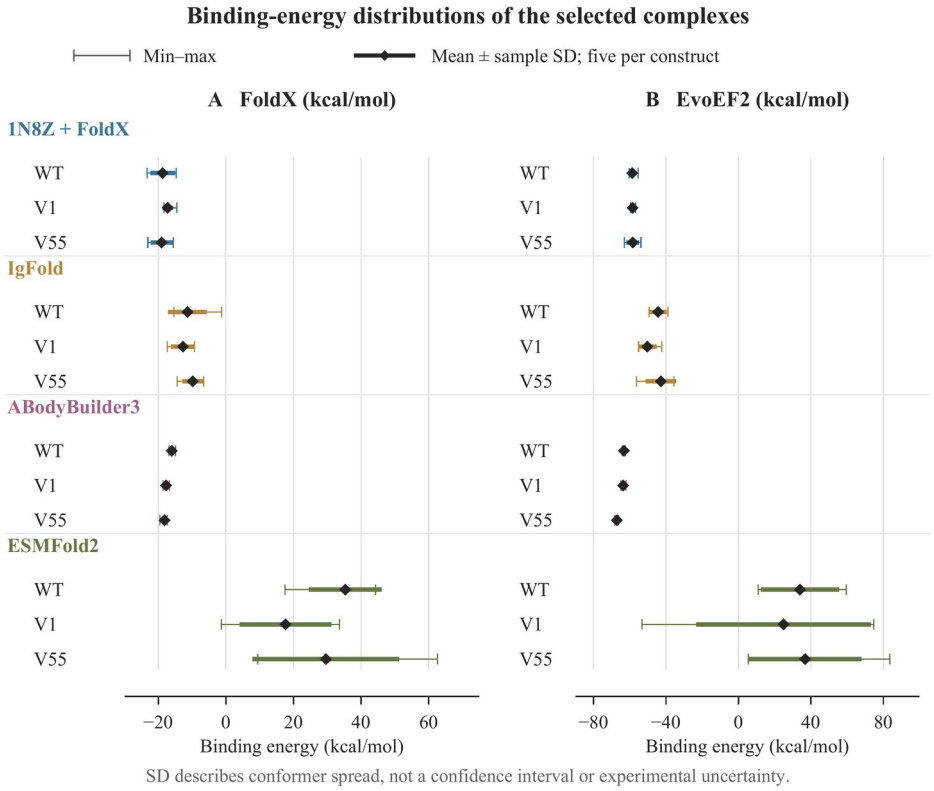


Figure A1. Binding-energy dispersion within each selected five-structure construct. Markers indicate means, thick intervals show mean ± sample SD, and thin capped intervals show observed minima and maxima. These are not confidence intervals. Both scorers use kcal mol^{-1}; conformers are not independent experimental measurements. ESMFold2 V1 includes exploratory seed9.

Table A10. FoldX construct energy summaries in kcal mol^{-1}.

Structure route	Variant	n	Mean	Sample SD	Observed range
1N8Z + FoldX	WT	5	−18.73	3.70	−23.35 to −14.72
1N8Z + FoldX	V1	5	−17.25	1.54	−18.34 to −14.53
1N8Z + FoldX	V55	5	−19.04	3.24	−23.09 to −15.61
ESMFold2	WT	5	35.34	10.81	17.49 to 44.28
ESMFold2	V1	5	17.63	13.60	−1.38 to 33.58
ESMFold2	V55	5	29.60	21.72	9.46 to 62.55
IgFold	WT	5	−11.36	5.78	−15.39 to −1.25

Structure route	Variant	n	Mean	Sample SD	Observed range
IgFold	V1	5	−12.72	3.47	−17.40 to −9.30
IgFold	V55	5	−9.83	3.04	−14.42 to −6.60
ABodyBuilder3	WT	5	−16.04	0.80	−16.91 to −14.95
ABodyBuilder3	V1	5	−17.68	0.83	−18.60 to −16.70
ABodyBuilder3	V55	5	−18.17	0.90	−19.58 to −17.34

The denominator for sample SD is n − 1. Means are equal-weight summaries of the selected structures; they are not Boltzmann-weighted free energies. Values are computed from unrounded source scores.

Table A11. EvoEF2 construct energy summaries in kcal mol^{-1}.

Structure route	Variant	n	Mean	Sample SD	Observed range
1N8Z + FoldX	WT	5	−58.63	1.94	−60.37 to −55.43
1N8Z + FoldX	V1	5	−58.42	1.18	−59.50 to −56.87
1N8Z + FoldX	V55	5	−58.36	3.56	−62.95 to −53.86
ESMFold2	WT	5	33.94	21.63	10.92 to 59.46
ESMFold2	V1	5	24.87	48.21	−53.24 to 74.76
ESMFold2	V55	5	36.86	31.13	5.51 to 83.61
IgFold	WT	5	−44.37	4.75	−49.34 to −38.97
IgFold	V1	5	−50.31	5.21	−55.09 to −42.31
IgFold	V55	5	−42.78	8.55	−56.25 to −35.63
ABodyBuilder3	WT	5	−63.19	0.69	−64.15 to −62.50
ABodyBuilder3	V1	5	−63.63	1.25	−64.89 to −62.08
ABodyBuilder3	V55	5	−67.13	0.88	−68.25 to −66.24

The denominator for sample SD is n − 1. Means are equal-weight summaries of the selected structures; they are not Boltzmann-weighted free energies. Values are computed from unrounded source scores.

For each route and scorer, $\Delta\Delta G_{pred}$ is the arithmetic mean of the five selected variant energies minus the arithmetic mean of the same route's five selected WT energies; no parent or seed pairing is used. Conditional affinities are then defined as $K_{D,variant}$ = 0.40 nM × exp($\Delta\Delta G_{pred}$/RT) at 298.15 K, with displayed RT = 0.59 kcal mol^{-1}. The full-precision RT constant is retained in the machine-readable companion and is used without display rounding in every calculation. Table A12 lists the group-mean energy differences and conditional transformations. A negative signed $\log_{10}$ error denotes a lower K_D (stronger affinity) than experiment. WT is the shared anchor and is excluded from error assessment. No calibration is fitted to V1 or V55.

Table A12. Group-mean binding changes and conditional K_D comparison.

Route and variant	Scorer	$\Delta\Delta G_{pred}$ (kcal mol^{-1})	Conditional K_D (nM)	Signed $\log_{10}$ error
1N8Z + FoldX V1	FoldX	1.48	4.88	1.17
1N8Z + FoldX V1	EvoEF2	0.20	0.56	0.23
1N8Z + FoldX V55	FoldX	−0.30	0.24	−1.88
1N8Z + FoldX V55	EvoEF2	0.27	0.63	−1.46
ESMFold2 V1	FoldX	−17.71	4.18×10^{-14}	−12.90
ESMFold2 V1	EvoEF2	−9.06	9.08×10^{-8}	−6.56
ESMFold2 V55	FoldX	−5.75	2.45×10^{-5}	−5.87
ESMFold2 V55	EvoEF2	2.92	55.08	0.48
IgFold V1	FoldX	−1.36	4.05×10^{-2}	−0.91
IgFold V1	EvoEF2	−5.94	1.76×10^{-5}	−4.27
IgFold V55	FoldX	1.54	5.35	−0.54

Route and variant	Scorer	$\Delta\Delta G_{pred}$ (kcal mol ^{1})	Conditional K_D (nM)	Signed $\log_{10}$ error
IgFold V55	EvoEF2	1.58	5.80	−0.50
ABodyBuilder3 V1	FoldX	−1.65	2.48×10^{-2}	−1.12
ABodyBuilder3 V1	EvoEF2	−0.44	0.19	−0.24
ABodyBuilder3 V55	FoldX	−2.13	1.09×10^{-2}	−3.22
ABodyBuilder3 V55	EvoEF2	−3.93	5.23×10^{-4}	−4.54

Each contrast uses five variant and five WT structures. Experimental K_D is 0.33 nM for V1 and 18.34 nM for V55 [5]. Both methods assume a unit-scale energy-to-free-energy mapping; these are conditional calculations, not independently calibrated affinities.

A6 Scorer association and strict tier sensitivity

Table A13. Descriptive association between FoldX and EvoEF2 binding scores.

Structure route	Construct	n structures	Pearson r	Spearman ρ
1N8Z + FoldX	All constructs	15	0.63	0.68
1N8Z + FoldX	WT	5	0.85	0.90
1N8Z + FoldX	V1	5	0.32	0.00
1N8Z + FoldX	V55	5	0.66	0.90
ESMFold2	All constructs	15	0.70	0.68
ESMFold2	WT	5	0.75	0.70
ESMFold2	V1	5	0.60	0.40
ESMFold2	V55	5	0.98	0.90
IgFold	All constructs	15	0.22	0.20
IgFold	WT	5	0.70	0.70
IgFold	V1	5	−0.36	0.00
IgFold	V55	5	−0.19	−0.30
ABodyBuilder3	All constructs	15	0.58	0.57
ABodyBuilder3	WT	5	0.34	0.40
ABodyBuilder3	V1	5	0.26	0.30
ABodyBuilder3	V55	5	0.15	−0.10

Both scores come from the same selected complex. All-construct coefficients pool 15 conformers within one route; construct-specific coefficients use five. No p-values, pooled-route coefficient or experimental-affinity correlation is reported. Dependence among conformers and effects of construct aggregation prevent interpreting these coefficients as independent cross-validation.

Table A14. ESMFold2 Variant 1 sensitivity to exploratory seed9.

Scorer	V1 cohort	n V1/WT	$\Delta\Delta G_{pred}$ (kcal mol ^{1})	Conditional K_D (nM)	Signed $\log_{10}$ error
FoldX	Selected five	5/5	−17.71	4.18×10^{-14}	−12.90
FoldX	Strict only	4/5	−12.96	1.27×10^{-10}	−9.41
EvoEF2	Selected five	5/5	−9.06	9.08×10^{-8}	−6.56
EvoEF2	Strict only	4/5	10.46	1.87×10^{7}	7.75

Strict-only analysis excludes V1 seed9; all five WT and all five V55 structures remain unchanged. EvoEF2 changes sign from favorable to unfavorable; FoldX remains favorable. This sensitivity is not a replacement of the fixed five-structure report cohort or a confidence interval.

Reproducibility of the arithmetic

The original companion includes numeric_analysis.py, numeric_analysis.json, selected_samples.csv, group_stats.csv, conditional_kd.csv, scorer_correlations.csv and strict_esm_v1.csv, together with four five_sample_mean_analysis.json files and source hashes. Full-precision per-model structural RMSDs, 12 group summaries, residue maps and coordinate

hashes accompany Appendix A7. The second-revision script recompute_raw_sensitivity.py independently recalculates the source-precision group statistics, conditional affinities and all 160 omissions, and compares them with the report-precision calculation. Source-precision group summaries and conditional affinities reconcile with the original companion at floating-point tolerance. Display rounding follows calculation. No structure prediction, refinement, docking, energy scoring or molecular-dynamics job was rerun.

A7 Structural visualization and reference-aligned heavy-chain comparison

Representative four-route WT structures

The first WT identifier in each route's saved five-structure selection list was resolved to its archived complex.pdb. Chain identity, sequence and contiguous sequential numbering were verified (H, 120 residues; L, 107 residues; HER2/A, 607 residues). For visualization only, the 607 matched HER2 Cα atoms were least-squares fitted to the FoldX-route HER2 coordinates and the resulting rigid transform was applied to every atom of each complete complex. A shared orthographic camera and scale were used. No antibody chain was repositioned independently, and no structure prediction, refinement, docking, energy scoring or molecular-dynamics job was rerun.

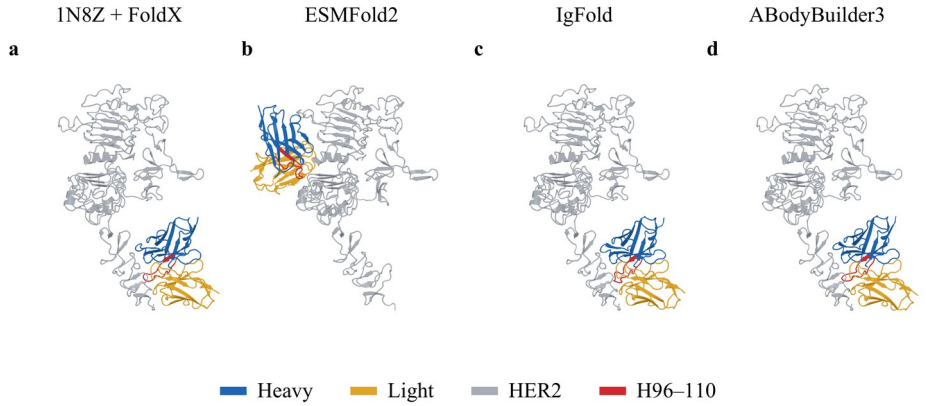


Figure A2. Representative WT trastuzumab variable-region–HER2 complexes from four modeling routes: (a) 1N8Z + FoldX, (b) ESMFold2, (c) IgFold, and (d) ABodyBuilder3. The heavy chain, light chain and HER2 are shown as blue, gold and gray cartoons, respectively; heavy-chain residues 96–110 are red (sequential numbering; CDR-H3 region including the conserved flanking Cys and Trp). One WT model per route is displayed, using the first entry in each saved analysis list. HER2 Cα atoms were used for whole-complex rigid alignment solely to obtain a common view and scale. ESMFold2 retains its jointly predicted interchain arrangement. The images illustrate saved models and do not establish epitope identity, binding affinity or model accuracy.

Figure A2 provenance: PNG SHA-256 481b9e01e0b093f93424f12a7e1607ef1b657883c7a05225381877ae1c96e83b; figure_manifest.json SHA-256 bb6d05b3f4b0f5ea771dd2e7bbdec544d7c9104699453ee3f95872c5b54c150b. Displayed record IDs: foldx-WT-V1-refined-run1-s9103000; esmfold2-WT-seed7; igfold-WT-refined-igfold_1-s9103000; abodybuilder3-WT-refined-plddt-loss-s9103000. The manifest retains source-coordinate hashes, PyMOL 3.1.0 identity, camera, rigid transforms, render settings and output hashes.

Heavy-chain comparison with the separately provided 1N8Z reference

For each of the fixed 60 selected complexes, model-chain H Cα atoms were paired by sequential position with the first 120 residues of chain B in the separately provided, hash-matched 1N8Z reference. A Kabsch least-squares fit used the 105 paired positions outside H96–110 (H1–95 and H111–120). The fitted coordinates were then used to calculate RMSD for those same 105 nonsegment positions and, without refitting, for the 15 H96–110 positions. The nonsegment mask

includes other CDR residues and therefore is not labeled a pure framework metric. Each route-by-construct summary is the arithmetic mean, sample SD and observed range of five fixed models. These reference-coordinate comparisons do not measure mutant error against a mutant crystal structure and do not establish native complex accuracy or epitope recovery.

Table A15. Heavy-chain RMSD summaries after fitting positions outside H96–110.

Structure route	Variant	n	Nonsegment mean ± SD (Å)	Nonsegment range (Å)	H96–110 mean ± SD (Å)	H96–110 range (Å)
1N8Z + FoldX	WT	5	$0.19 \pm 1.03 \times 10^{-3}$	0.19–0.19	$0.25 \pm 6.82 \times 10^{-3}$	0.24–0.26
1N8Z + FoldX	V1	5	$0.20 \pm 1.63 \times 10^{-3}$	0.19–0.20	0.42 ± 0.03	0.37–0.44
1N8Z + FoldX	V55	5	$0.20 \pm 8.93 \times 10^{-3}$	0.19–0.21	0.28 ± 0.02	0.26–0.30
ESMFold2	WT	5	0.42 ± 0.02	0.41–0.44	2.44 ± 0.16	2.18–2.61
ESMFold2	V1	5	0.41 ± 0.04	0.38–0.48	2.38 ± 0.74	1.25–3.07
ESMFold2	V55	5	0.41 ± 0.02	0.39–0.44	2.02 ± 0.30	1.73–2.52
IgFold	WT	5	0.57 ± 0.03	0.54–0.61	1.51 ± 0.11	1.38–1.67
IgFold	V1	5	0.59 ± 0.01	0.58–0.61	2.39 ± 0.57	1.79–2.97
IgFold	V55	5	0.59 ± 0.02	0.58–0.62	3.12 ± 0.62	2.43–3.92
ABodyBuilder3	WT	5	0.48 ± 0.02	0.46–0.50	0.77 ± 0.03	0.74–0.80
ABodyBuilder3	V1	5	$0.49 \pm 7.08 \times 10^{-3}$	0.48–0.49	1.13 ± 0.05	1.06–1.18
ABodyBuilder3	V55	5	0.48 ± 0.02	0.45–0.50	0.93 ± 0.06	0.88–1.00

The nonsegment RMSD is calculated on the 105 Cα positions used for fitting; the H96–110 RMSD is calculated on the 15 excluded positions after that same fit. SD is the sample SD with denominator n − 1. Values are computed from unrounded per-model coordinates and displayed to two decimal places; ranges are observed minima and maxima, not confidence intervals.

Table A16. Per-model heavy-chain RMSD values relative to 1N8Z.

Structure route	Variant	Record ID	Fit Cα n	Nonsegment RMSD (Å)	H96–110 RMSD (Å)
1N8Z + FoldX	WT	R01	105	0.19	0.26
1N8Z + FoldX	WT	R03	105	0.19	0.26
1N8Z + FoldX	WT	R05	105	0.19	0.24
1N8Z + FoldX	WT	R07	105	0.19	0.25
1N8Z + FoldX	WT	R09	105	0.19	0.26
1N8Z + FoldX	V1	R02	105	0.19	0.43
1N8Z + FoldX	V1	R06	105	0.20	0.44
1N8Z + FoldX	V1	R10	105	0.20	0.43
1N8Z + FoldX	V1	R14	105	0.20	0.37
1N8Z + FoldX	V1	R18	105	0.20	0.42
1N8Z + FoldX	V55	R04	105	0.19	0.27
1N8Z + FoldX	V55	R08	105	0.21	0.29
1N8Z + FoldX	V55	R12	105	0.20	0.30
1N8Z + FoldX	V55	R16	105	0.20	0.26
1N8Z + FoldX	V55	R20	105	0.19	0.28
ESMFold2	WT	WT s7	105	0.44	2.49
ESMFold2	WT	WT s15	105	0.41	2.51
ESMFold2	WT	WT s1	105	0.42	2.61
ESMFold2	WT	WT s14	105	0.44	2.42
ESMFold2	WT	WT s2	105	0.42	2.18
ESMFold2	V1	V1 s3	105	0.42	2.13
ESMFold2	V1	V1 s14	105	0.48	1.25

Structure route	Variant	Record ID	Fit Cα n	Nonsegment RMSD (Å)	H96–110 RMSD (Å)
ESMFold2	V1	V1 s16	105	0.39	3.07
ESMFold2	V1	V1 s15	105	0.39	2.48
ESMFold2	V1	V1 s9	105	0.38	2.98
ESMFold2	V55	V55 s0	105	0.41	2.02
ESMFold2	V55	V55 s11	105	0.41	1.90
ESMFold2	V55	V55 s1	105	0.44	2.52
ESMFold2	V55	V55 s12	105	0.40	1.73
ESMFold2	V55	V55 s16	105	0.39	1.92
IgFold	WT	R01	105	0.57	1.55
IgFold	WT	R04	105	0.58	1.38
IgFold	WT	R07	105	0.54	1.67
IgFold	WT	R10	105	0.61	1.43
IgFold	WT	R13	105	0.57	1.54
IgFold	V1	R02	105	0.61	2.96
IgFold	V1	R05	105	0.58	2.37
IgFold	V1	R08	105	0.59	1.79
IgFold	V1	R11	105	0.58	1.85
IgFold	V1	R14	105	0.60	2.97
IgFold	V55	R03	105	0.58	3.36
IgFold	V55	R06	105	0.62	3.92
IgFold	V55	R09	105	0.58	2.43
IgFold	V55	R12	105	0.58	2.54
IgFold	V55	R15	105	0.59	3.34
ABodyBuilder3	WT	R01	105	0.47	0.80
ABodyBuilder3	WT	R04	105	0.50	0.74
ABodyBuilder3	WT	R07	105	0.47	0.79
ABodyBuilder3	WT	R13	105	0.46	0.79
ABodyBuilder3	WT	R16	105	0.50	0.76
ABodyBuilder3	V1	R02	105	0.48	1.16
ABodyBuilder3	V1	R05	105	0.49	1.10
ABodyBuilder3	V1	R08	105	0.48	1.18
ABodyBuilder3	V1	R11	105	0.49	1.06
ABodyBuilder3	V1	R14	105	0.48	1.15
ABodyBuilder3	V55	R03	105	0.45	0.99
ABodyBuilder3	V55	R06	105	0.50	0.88
ABodyBuilder3	V55	R09	105	0.45	1.00
ABodyBuilder3	V55	R12	105	0.49	0.89
ABodyBuilder3	V55	R18	105	0.49	0.88

Short record labels match Tables A6–A9. Full record IDs, coordinate paths and hashes, residue maps, Kabsch/quaternion agreement checks and full-precision values are retained in structural_rmsd_records.json (SHA-256 14d11818d8eca0c9c71989ec194336dd8fea8b3ad7353a8605d1cf34c2831f4b); full-precision group statistics are retained in structural_rmsd_summary.json (SHA-256 6b6662310303af312f6072be0dfcd95c2b4b5cb5fb3ce1c1c254a7211b3ec7ff). The exact R and RT constants and all full-precision energy-derived calculations remain in numeric_analysis.json. Separately provided 1N8Z reference SHA-256: ed934d36503335c6764b08880f9e5093 3d99eb13b9bf145339cfcaee150eb21e.

A8 Conformer exclusion sensitivity

For each route, scorer and variant, we recomputed the difference of arithmetic means after each of five variant omissions (4 variant/5 WT) and each of five WT omissions (5 variant/4 WT). These ten deletions are deterministic

diagnostics of the selected cohort, not independent replicates, confidence intervals or thermodynamic ensembles. No parent pairing, resampling, rescoring or new structure generation was performed.

Primary inputs are the retained per-record source scores and selected IDs, checked against the original evidence and analysis companion. The original report-precision supplement remains a separate comparison. Relative to its four-decimal inputs, the maximum absolute change is 0.000008 kcal mol^{-1} for baseline differences and 0.000012 kcal mol^{-1} for omission differences; the corresponding maximum changes in signed $\log_{10}$ affinity error are 5.86×10^{-6} and 8.80×10^{-6}. No baseline or omission sign classification changes. Thus recovering source precision confirms the first revision's sensitivity result rather than correcting its seven-of-sixteen count.

Table A17. Single-conformer omission ranges in kcal mol^{-1}.

Route and variant	Scorer	Baseline	Omission range	Sign retained in all 10
1N8Z + FoldX V1	FoldX	+1.48	+0.33 to +2.49	Yes
1N8Z + FoldX V1	EvoEF2	+0.20	−0.23 to +1.00	No
1N8Z + FoldX V55	FoldX	−0.30	−1.46 to +0.71	No
1N8Z + FoldX V55	EvoEF2	+0.27	−0.85 to +1.42	No
ESMFold2 V1	FoldX	−17.71	−22.17 to −12.96	Yes
ESMFold2 V1	EvoEF2	−9.06	−21.54 to +10.46	No
ESMFold2 V55	FoldX	−5.75	−13.99 to −0.71	Yes
ESMFold2 V55	EvoEF2	+2.92	−8.77 to +10.75	No
IgFold V1	FoldX	−1.36	−2.36 to +1.17	No
IgFold V1	EvoEF2	−5.94	−7.94 to −4.59	Yes
IgFold V55	FoldX	+1.54	+0.53 to +4.06	Yes
IgFold V55	EvoEF2	+1.58	−0.20 to +4.95	No
ABodyBuilder3 V1	FoldX	−1.65	−1.89 to −1.37	Yes
ABodyBuilder3 V1	EvoEF2	−0.44	−0.82 to −0.12	Yes
ABodyBuilder3 V55	FoldX	−2.13	−2.35 to −1.78	Yes
ABodyBuilder3 V55	EvoEF2	−3.93	−4.17 to −3.65	Yes

Sign retention concerns $\Delta\Delta G_{pred}$ relative to WT, not accuracy. Seven of 16 contrasts reverse sign in 19 of 160 deletions. Signed $\log_{10}$ affinity error uses the distinct threshold of experimental K_D; six contrasts cross it in 14 deletions. Each IgFold scorer retains V1 < WT < V55 after 14 of 15 unique deletions, failing for WT R04 (FoldX) or V55 R15 (EvoEF2). These fixed-cohort diagnostics are not confidence intervals. The supplement contains all 160 source-precision and report-precision results, IDs, hashes and recomputation code. Four-decimal input rounding permits at most ±0.0001 kcal mol^{-1} per mean difference; the nearest omission value is +0.0030, leaving signs resolved.

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